MTH2010 Algebra

Problem Sheet 2

For tutorials in Week 7 questions 7 and 11 will be discussed.

1. Find all conjugacy classes and their cardinalities in the following groups:

(a) D_8,

(b) A_4,

(c) $\mathbb{Z}/2\mathbb{Z} \times S_3$.

In each case verify the class equation and find the centre of the group.

2. Let G be a finite group and $Z(G)$ its centre. Assume that

$$|G : Z(G)| = n.$$

Prove that every conjugacy class in G has at most n elements.

3. **(a)** Let $\alpha = (123)(4567) \in S_{10}$. Compute the cardinality of $\mathrm{orb}(\alpha)$ under the conjugation action.

(b) With α as above, express the centraliser $C_{S_{10}}(\alpha)$ of α as a set of products of powers of cycles and permutations, as in Section 4.3 of the lecture notes.

(c) Let $\sigma = (123)(456)(78) \in S_{10}$. Compute the cardinality of $\mathrm{orb}(\sigma)$, the orbit of σ under the conjugation action.

4. Show that if n is odd then the n-cycles of the symmetric group S_n form two conjugacy classes of equal size in A_n.

5. Let G be a finite group whose cardinality is an odd positive integer. Let x be an element of G distinct from e_G. Prove that x and x^{-1} are not conjugate in G.

6. Using the usual generators r, s, and relation $sr^i = r^{-i}s = r^{n-i}s$ for the dihedral group D_{2n} show the following.

(a) For $n = 2k$ even, the conjugacy classes in D_{2n} are:

$$\{1\}, \{r^k\}, \{r, r^{-1}\}, \{r^2, r^{-2}\}, \ldots, \{r^{k-1}, r^{1-k}\}, \{sr^{2j} : 1 \leq j \leq k\}, \{sr^{2j-1} : 1 \leq j \leq k\}.$$

Write the class equation for D_{2n} in this case.

(b) For $n = 2k + 1$ odd, the conjugacy classes in D_{2n} are:

$$\{1\}, \{r, r^{-1}\}, \{r^2, r^{-2}\}, \ldots, \{r^k, r^{-k}\}, \{sr^j : 1 \leq j \leq n\}, \{sr^j : 1 \leq j \leq n\}.$$

Write the class equation for D_{2n} in this case.

7. Let p be a prime number and let S_p be the symmetric group of permutations of the set $\{1, 2, \ldots, p\}$. Prove that if P is a subgroup of S_p of cardinality p then

$$|N_{S_p}(P)| = p(p-1).$$

[Hint: compute the cardinality of the set $\mathrm{Sub}_p(S_p)$ of subgroups of cardinality p of S_p and show that S_p acts transitively by conjugation on $\mathrm{Sub}_p(S_p)$.]

8. **(a)** Find all Sylow 2-subgroups of S_4. If $n_2(S_4) > 1$ find elements of S_4 which conjugate one of these Sylow 2-subgroup into each of the others.

(b) Find two distinct Sylow 2-subgroups of S_5 and an element of S_5 which conjugates one into the other.

9. Prove that a group of cardinality 56, or cardinality 312, has a normal p-subgroup for some prime p dividing its order.

10. **(a)** Find $n_5(S_5)$ and $n_5(A_5)$.

(b) Find $n_2(S_5)$.

(c) Find $n_2(A_5)$.
Here $n_p(G)$ denotes the number of Sylow p-subgroups of a finite group G. [Note that A_4 can be viewed as the subgroup of A_5 of those permutations fixing an $i \in \{1, 2, 3, 4, 5\}$.]

11. Let p be a prime number and let G be a finite group whose cardinality is divisible by p. Let P be a Sylow p-subgroup of G and let H be a subgroup of G whose cardinality is divisible by p.

(a) Assume that P is a normal subgroup of G. Prove that $P \cap H$ is the unique Sylow p-subgroup of H.

(b) Prove that $(gPg^{-1}) \cap H$ is a Sylow p–subgroup of H for some $g \in G$, whether or not P is normal in G.